

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("Titanic-Dataset fulldata.csv")
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	C
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	

Cabin	Embarked
NaN	S
C85	C
NaN	S
C123	S
NaN	S

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype  
---  -
 0   PassengerId     891 non-null   int64  
 1   Survived        891 non-null   int64  
 2   Pclass          891 non-null   int64  
 3   Name            891 non-null   object  
 4   Sex             891 non-null   object  
 5   Age             714 non-null   float64 
 6   SibSp           891 non-null   int64  
 7   Parch           891 non-null   int64  
 8   Ticket          891 non-null   object  
 9   Fare            891 non-null   float64 
10   Cabin           204 non-null   object  
11   Embarked        889 non-null   object  
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
df.describe()
```

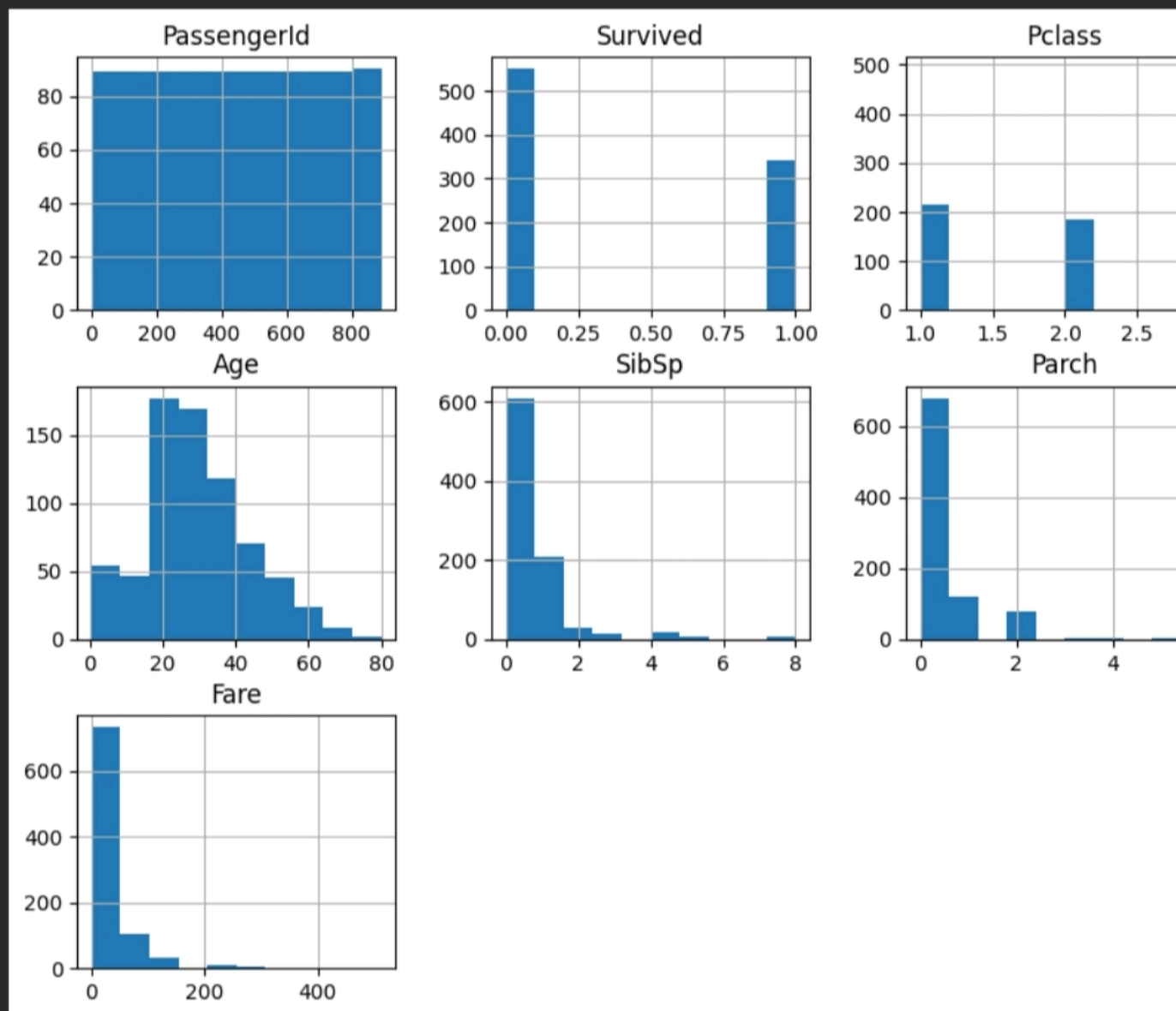
	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
df.isnull().sum()
```

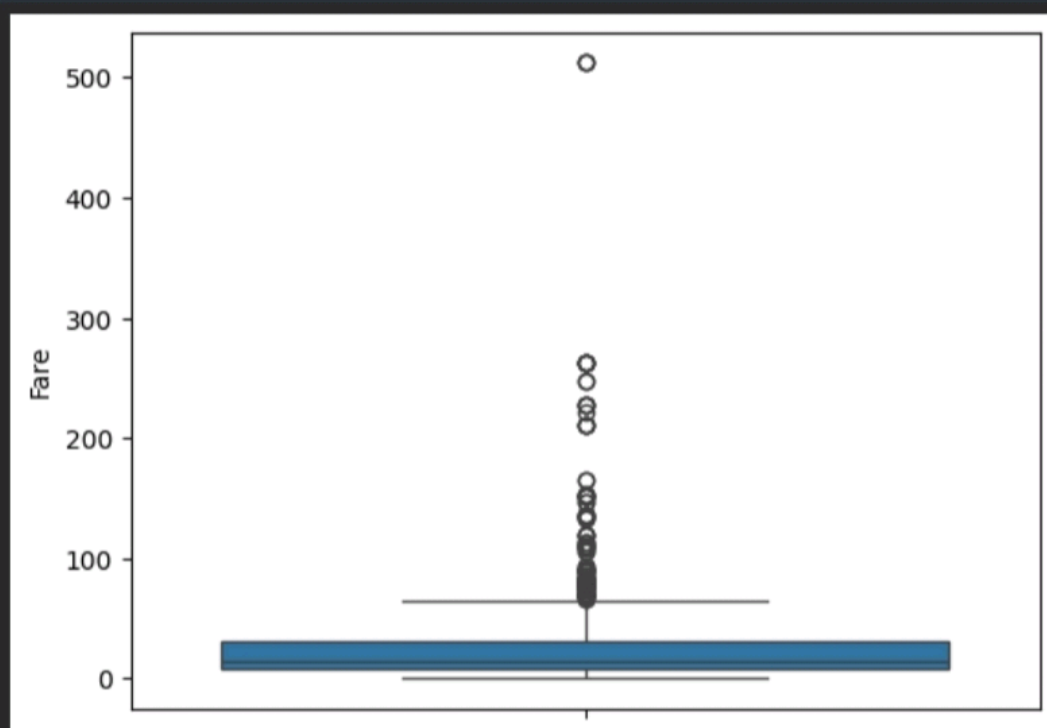
	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

```
df.hist(figsize=(10,8))
plt.show()
```

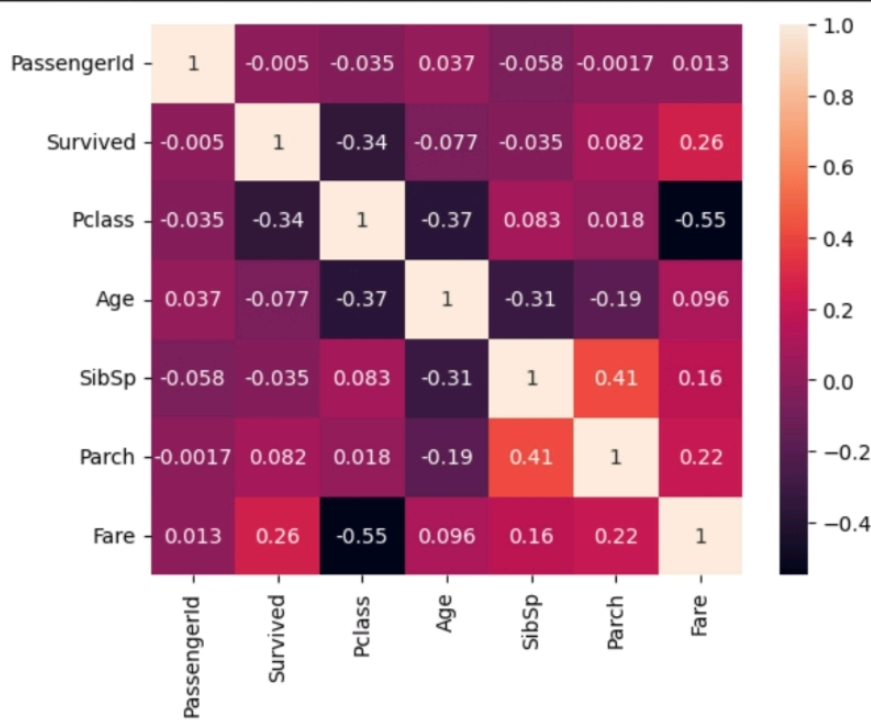


```
sns.boxplot(y=df["Fare"])
plt.show()
```



```
numeric_df = df.select_dtypes(include=['number'])
```

```
sns.heatmap(numeric_df.corr(), annot=True)  
plt.show()
```



```
sns.pairplot(df[['Age', 'Fare', 'Pclass', 'Survived']])  
plt.show()
```

